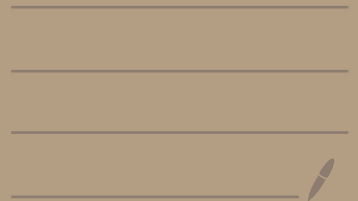


Lec. 8. Existence and Uniqueness.

· Continuous dependence.

- Sec. 2-8 [BDM] [easier presentation. Longer proof].
- Sec 2.2. [T] Reference.
Sec 2-4 [T].



Consider 1st order linear system.

$$\frac{d}{dt} X(t) = F(t, X(t)), \quad X \in \mathbb{R}^{n \times 1}, F \in \mathbb{R}^{n \times 1}.$$

$$\begin{cases} X(t_0) = X_0. \end{cases}$$

Q: When there exists a solution?

We will focus on $n=1$. The same statement holds for $n \geq 1$, but require technical treatment i.e. vector norms

$$n=1, \quad F \mapsto f.$$

$$\frac{d}{dt} y(t) = f(t, y) \quad y(t_0) = y_0, \quad (1)$$

Lipschitz function: $f(x)$ is Lipschitz in y

$$\text{if } \exists L > 0 \text{ s.t. } |f(x) - f(y)| \leq L \cdot |x - y|.$$

Thus: [Variant of Thm 2.1].



If $f(t, y)$ is continuous in $|t - t_0| \leq a, |y - y_0| \leq b$ Ass 1

and $f(t, \cdot)$ is Lipschitz in $\mathbb{R}$. Then Ass 2

$\exists 0 < h \leq a$ s.t. there exists a unique solution $y(t)$ for $|t - t_0| \leq h$.

$$\text{Ass 1} \Rightarrow |f(t, y)| \leq M \quad (t, y) \in \mathbb{R}^2: [-a, 0] \times [t_0, t_0 + b]$$

$$\text{Ass 2} \Rightarrow |f(t, y_1) - f(t, y_2)| \leq L \cdot |y_1 - y_2|.$$

Pf: ① rewrite ① as integral form. $y = \phi(t)$

$$\Rightarrow \phi(t) = y_0 + \int_{t_0}^t f(s, \phi(s)) ds$$

② Successive iterate $\phi_0 = y_0$ $I_y = [y_0 - h, y_0 + h]$

$$\phi_{n+1} = y_0 + \int_{t_0}^t f(s, \phi_n(s)) ds.$$

• If $\phi_n \in I_y$, $\Rightarrow \phi_{n+1} \in I_y$

$$|\phi_{n+1} - y_0| \leq M \cdot t \leq Mh$$

$$\text{• If } h < \frac{a}{M} \Rightarrow |\phi_{n+1} - y_0| \leq b.$$

③ Good. show convergence for small h .

$$|\phi_1 - \phi_0| = |\phi_1 - y_0| \leq M \cdot h.$$

Use induction to show $|\phi_{n+1} - \phi_n| \leq (Lh)^n \cdot M \cdot h.$ $n \geq 0$

$$\phi_{n+1} - \phi_n = \int_{t_0}^t f(s, \phi_n(s)) - f(s, \phi_{n-1}(s)) ds.$$

$$\Rightarrow |\phi_{n+1} - \phi_n| \leq \int_0^t L |\phi_n(s) - \phi_{n-1}(s)| ds.$$

$$\leq L \cdot \int_0^t (Lh)^{n-1} \cdot Mh.$$

$$\leq (Lh)^n \cdot Mh$$

Now take $Lh < 1$.

Then

$$\phi(t) = \sum_{n \geq 1} (\phi_{n+1} - \phi_n) + y_0, \text{ converge.}$$

$$|\phi_n - \phi(t)| \leq \sum_{k=0}^n (Lh)^k \cdot nh$$

$$= \frac{(Lh)^{n+1}}{1-Lh} \cdot nh \rightarrow 0$$

$$\phi_{n+1}(t) = \int_0^t f(s, \phi_n(s)) ds$$

send $n \rightarrow \infty$

$$\phi(t) = \int_0^t f(s, \phi(s)) ds \quad \square$$

• Show continuous and ϕ is C^1 .

• Uniqueness: Compare 2 solutions.

$$\phi(t) = y_0 + \int_0^t f(s, \phi(s)) ds$$

$$\psi(t) = y_0 + \int_0^t f(s, \psi(s)) ds.$$

estimate

$$|\phi(t) - \psi(t)| \leq \int_0^t L |\phi(s) - \psi(s)| ds.$$

Solve the differential inequality.

Remark .. Existence of ODE solution

. Only requires f to be continuous.

. Lipschitz condition

$$|f(s, y_1) - f(s, y_2)| \leq L \cdot |y_1 - y_2|.$$

is necessary for uniqueness.

Example:

$$\begin{cases} \frac{d}{dt} x = x^{\frac{1}{2}} \\ x(0) = 0. \end{cases}, \quad f(s, x) = x^{\frac{1}{2}}$$

Q: come up
2 solutions

. $x_1(t) \equiv 0$

. $x_2(t) \equiv t^2$ 2nd solution. Not unique.

Not unique.